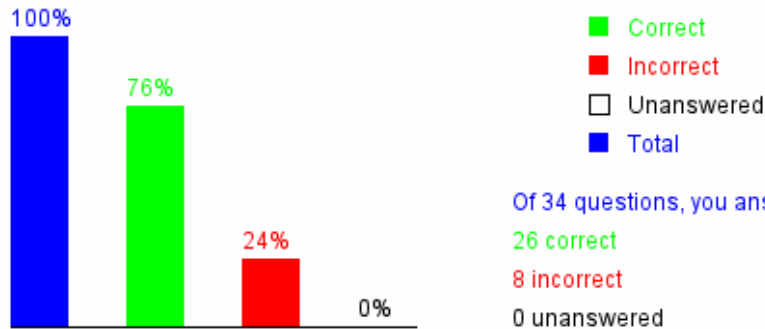


Introduction to Programming Using Python, Y. Daniel Liang

Many questions in this edition have been updated in the new edition. Please check with the publisher on the newest edition.

This quiz is for students to practice. A large number of additional quiz is available for instructors using Quiz Generator from the Instructor's Resource Website. Videos for Java, Python, and C++ can be found at <https://yongdanielliang.github.io/revelvideos.html>.

Chapter 4 Selections



Of 34 questions, you answered:

26 correct

8 incorrect

0 unanswered

Please send suggestions and errata to Dr. Liang at y.daniel.liang@gmail.com. Indicate which book and edition you are using. Thanks!

Section 4.2 Boolean Types, Values, and Expressions

4.1 The "less than or equal to" comparison operator is _____.

- ☐ A. <
- ☒ B. <=
- ☐ C. =<
- ☐ D. <<
- ☐ E. !=

Your answer is correct



4.2 The equal comparison operator is _____.

- ☐ A. <>
- ☐ B. !=
- ☒ C. ==
- ☐ D. =

Your answer is correct



4.3 The word True is _____.

- ☒ A. a Python keyword
- ☒ B. a Boolean literal
- ☒ C. same as value 1
- ☐ D. same as value 0

Your answer ABC is incorrect



Hint: True is a Boolean literal just like integer literal 10. True is also a keyword

[Click here to show the correct answer](#)

Section 4.3 Generating Random Numbers

4.4 To generate a random integer between 0 and 5, use _____.

- ☐ A. `random.randint(0, 5)`

- ☒ B. `random.randint(0, 6)`
☐ C. `random.randrange(0, 5)`
☐ D. `random.randrange(0, 6)`

Your answer B is incorrect



[Click here to show the correct answer](#)

4.5 `random.randint(0, 1)` returns _____.

- ☒ A. 0
☐ B. 1
☐ C. 0 or 1
☐ D. 2

Your answer A is incorrect



[Click here to show the correct answer](#)

4.6 `random.random()` returns _____.

- ☐ A. a float number `i` such that $0 < i < 1.0$
☒ B. a float number `i` such that $0 \leq i < 1.0$
☐ C. a float number `i` such that $0 \leq i \leq 1.0$
☐ D. a float number `i` such that $0 < i < 2.0$

Your answer is correct



Sections 4.4-4.10

4.7 Which of the following code displays the area of a circle if the radius is positive.

- ☐ A. `if radius != 0: print(radius * radius * 3.14159)`
☐ B. `if radius >= 0: print(radius * radius * 3.14159)`
☒ C. `if radius > 0: print(radius * radius * 3.14159)`
☐ D. `if radius <= 0: print(radius * radius * 3.14159)`

Your answer is correct



4.8 What is the output of the following code?

```
x = 0
if x < 4:
    x = x + 1

print("x is", x)
```

- ☐ A. x is 0
☒ B. x is 1
☐ C. x is 2
☐ D. x is 3
☐ E. x is 4

Your answer is correct



4.9 Suppose `isPrime` is a boolean variable, which of the following is the correct and best statement for testing if `isPrime` is true.

- ☐ A. `if isPrime = True:`
☐ B. `if isPrime == True:`
☒ C. `if isPrime:`

- ☐ D. if not isPrime = False:
- ☐ E. if not isPrime == False:

Your answer is correct



4.10 Analyze the following code:

```
even = False
if even = True:
    print("It is even!")
```

- ☐ A. The program has a syntax error in line 1 (even = False)
- ☒ B. The program has a syntax error in line 2 if even = True is not a correct condition. It should be replaced by if even == True: or if even:.
- ☐ C. The program runs, but displays nothing.
- ☐ D. The program runs and displays It is even!.

Your answer is correct



4.11 Analyze the following code.

```
even = False
if even:
    print("It is even!")
```

- ☐ A. The code displays It is even!
- ☒ B. The code displays nothing.
- ☐ C. The code is wrong. You should replace if even: with if even == True:
- ☐ D. The code is wrong. You should replace if even: with if even = True:

Your answer is correct



4.12 Suppose x = 1, y = -1, and z = 1. What will be displayed by the following statement?

```
if x > 0:
    if y > 0:
        print("x > 0 and y > 0")
elif z > 0:
    print("x < 0 and z > 0")
```

- ☐ A. x > 0 and y > 0
- ☐ B. x < 0 and z > 0
- ☐ C. x < 0 and z < 0
- ☒ D. nothing displayed

Your answer is correct



4.13 The following code displays _____.

```
temperature = 50

if temperature >= 100:
    print("too hot")
elif temperature <= 40:
    print("too cold")
else:
    print("just right")
```

- ☐ A. too hot
- ☐ B. too cold
- ☒ C. just right

- ☐ D. too hot too cold just right

Your answer is correct



4.14 Analyze the following code:

Code 1:

```
if number % 2 == 0:
    even = True
else:
    even = False
```

Code 2:

```
even = number % 2 == 0
```

- ☐ A. Code 1 has compile errors.
- ☐ B. Code 2 has compile errors.
- ☐ C. Both Code 1 and Code 2 have compile errors.
- ☒ D. Both Code 1 and Code 2 are correct, but Code 2 is better.

Your answer is correct



4.15 Suppose income is 4001, what will be displayed by f the following code?

```
if income > 3000:
    print("Income is greater than 3000")
elif income > 4000:
    print("Income is greater than 4000")
```

- ☐ A. none
- ☒ B. Income is greater than 3000
- ☐ C. Income is greater than 3000 followed by Income is greater than 4000
- ☐ D. Income is greater than 4000
- ☐ E. Income is greater than 4000 followed by Income is greater than 3000

Your answer is correct



4.16 Suppose you write the code to display "Cannot get a driver's license" if age is less than 16

and "Can get a driver's license" if age is greater than or equal to 16. Which of the following code is correct?

I:

```
if age < 16:
    print("Cannot get a driver's license")
if age >= 16:
    print("Can get a driver's license")
```

II:

```
if age < 16:
    print("Cannot get a driver's license")
else:
    print("Can get a driver's license")
```

III:

```
if age < 16:
    print("Cannot get a driver's license")
elif age >= 16:
    print("Can get a driver's license")
```

IV:

```
if age < 16:  
    print("Cannot get a driver's license")  
elif age == 16:  
    print("Can get a driver's license")  
elif age > 16:  
    print("Can get a driver's license")
```

- ☐ A. I and II
☐ B. II and III
☐ C. I, II, and III
☐ D. III and IV
☒ E. All correct

Your answer is correct



4.17 Suppose you write the code to display "Cannot get a driver's license" if age is less than 16

and "Can get a driver's license" if age is greater than or equal to 16.
Which of the following code is the best?

I:

```
if age < 16:  
    print("Cannot get a driver's license")  
if age >= 16:  
    print("Can get a driver's license")
```

II:

```
if age < 16:  
    print("Cannot get a driver's license")  
else:  
    print("Can get a driver's license")
```

III:

```
if age < 16:  
    print("Cannot get a driver's license")  
elif age >= 16:  
    print("Can get a driver's license")
```

IV:

```
if age < 16:  
    print("Cannot get a driver's license")  
elif age == 16:  
    print("Can get a driver's license")  
elif age > 16:  
    print("Can get a driver's license")
```

- ☐ A. I
☐ B. II
☐ C. III
☒ D. IV

Your answer D is incorrect



[Click here to show the correct answer](#)

4.18 The _____ function immediately terminates the program.

- ☐ A. sys.terminate()
☐ B. sys.halt()
☒ C. sys.exit()
☐ D. sys.stop()

Your answer is correct 

Section 4.11 Logical Operators

4.19 Which of the Boolean expressions below is incorrect?

- ☐ A. $\text{True and } 3 \Rightarrow 4$
- ☒ B. $!(x > 0) \text{ and } (x > 0)$
- ☒ C. $(x > 0) \text{ or } (x < 0)$
- ☒ D. $(x \neq 0) \text{ or } (x = 0)$
- ☒ E. $(-10 < x < 0)$

Your answer BCDE is incorrect 

Hint: a: $(3 \Rightarrow 4)$ should be $(3 \geq 4)$, b: $!(x > 0)$ should be not $(x > 0)$ d: $(x = 0)$ should be $(x == 0)$

[Click here to show the correct answer](#)

4.20 Which of the following is the correct expression that evaluates to True if the number x is between 1 and 100 or the number is negative?

21. To check whether a char variable ch **is** an uppercase letter, you write _____.

- ☒ A. $(ch \geq 'A' \text{ and } ch \leq 'Z')$
- ☐ B. $(ch \geq 'A' \text{ and } ch \leq 'Z')$
- ☐ C. $(ch \geq 'A' \text{ or } ch \leq 'Z')$
- ☐ D. $('A' \leq ch \leq 'Z')$

Your answer A is incorrect 

[Click here to show the correct answer](#)

4.21 Given $|x - 2| \leq 4$, Which of the following is true?

- ☒ A. $x - 2 \leq 4 \text{ and } x - 2 \geq 4$
- ☐ B. $x - 2 \leq 4 \text{ and } x - 2 > -4$
- ☐ C. $x - 2 \leq 4 \text{ and } x - 2 \geq -4$
- ☐ D. $x - 2 \leq 4 \text{ or } x - 2 \geq -4$

Your answer A is incorrect 

[Click here to show the correct answer](#)

4.22 Given $|x - 2| \geq 4$, Which of the following is true?

- ☐ A. $x - 2 \geq 4 \text{ and } x - 2 \leq -4$
- ☒ B. $x - 2 \geq 4 \text{ or } x - 2 \leq -4$
- ☐ C. $x - 2 \geq 4 \text{ and } x - 2 < -4$
- ☐ D. $x - 2 \geq 4 \text{ or } x - 2 \leq -4$

Your answer is correct 

4.23 Assume $x = 4$ and $y = 5$, Which of the following is true?

- ☐ A. $x < 5 \text{ and } y < 5$
- ☒ B. $x < 5 \text{ or } y < 5$
- ☐ C. $x > 5 \text{ and } y > 5$
- ☐ D. $x > 5 \text{ or } y > 5$

Your answer is correct



4.24 Assume $x = 4$ and $y = 5$, Which of the following is true?

- ☐ A. `not (x == 4)`
- ☐ B. `x != 4`
- ☐ C. `x == 5`
- ☒ D. `x != 5`

Your answer is correct



4.25 Assume $x = 4$ and $y = 5$, Which of the following is true?

- ☐ A. `not (x == 4)`
- ☐ B. `x != 4`
- ☐ C. `x == 5`
- ☒ D. `x != 5`

Your answer is correct



4.26 Which of the following is equivalent to $x \neq y$?

- ☒ A. `not (x == y)`
- ☐ B. `x > y and x < y`
- ☒ C. `x > y or x < y`
- ☐ D. `x >= y or x <= y`

Your answer is correct



4.27 What will be displayed by the following code?

```
ch = 'F'
if ch >= 'A' and ch <= 'Z':
    print(ch)
```

- ☒ A. F
- ☐ B. f
- ☐ C. nothing
- ☐ D. F f

Your answer is correct



Section 4.14 Conditional Expressions

4.28 What is y after the following statement is executed?

```
x = 0
y = 10 if x > 0 else -10
```

- ☒ A. -10
- ☐ B. 0
- ☐ C. 10
- ☐ D. 20
- ☐ E. Illegal expression

Your answer is correct



4.29 Analyze the following code fragments that assign a boolean value to the variable even.

Code 1:
`if number % 2 == 0:
 even = True
else:
 even = False`

Code 2:
`even = True if number % 2 == 0 else False`

Code 3:
`even = number % 2 == 0`

- ☐ A. Code 2 has a syntax error, because you cannot have True and False literals in the conditional expression.
- ☐ B. Code 3 has a syntax error, because you attempt to assign number to even.
- ☐ C. All three are correct, but Code 1 is preferred.
- ☐ D. All three are correct, but Code 2 is preferred.
- ☒ E. All three are correct, but Code 3 is preferred.

Your answer is correct 
[Click here to show an explanation](#)

4.30 What will be displayed by the following code?

```
isCorrect = False  
print("Correct" if isCorrect else "Incorrect")
```

- ☐ A. Correct
- ☒ B. Incorrect
- ☐ C. nothing
- ☐ D. Correct Incorrect

Your answer is correct 

Section 4.15 Operator Precedence and Associativity

4.31 The order of the precedence (from high to low) of the operators +, *, and, or is:

- ☐ A. and, or, *, +
- ☐ B. *, +, and, or
- ☒ C. *, +, and, or
- ☐ D. *, +, or, and
- ☐ E. or, and, *, +

Your answer is correct 

4.32 Which of the following operators are right-associative.

- ☐ A. *
- ☐ B. +
- ☐ C. %
- ☐ D. and
- ☒ E. =

Your answer is correct 

4.33 What is the value of the following expression?

True or True and False

- ☒ A. True
☐ B. False

Your answer is correct



[Click here to show an explanation](#)

4.34 Which of the following statements are True?

- ☒ A. $(x > 0 \text{ and } x < 10)$ is same as $(x > 0 \text{ and } x < 10)$
☒ B. $(x > 0 \text{ or } x < 10)$ is same as $(0 < x < 10)$
☒ C. $(x > 0 \text{ or } x < 10 \text{ and } y < 0)$ is same as $(x > 0 \text{ or } (x < 10 \text{ and } y < 0))$
☒ D. $(x > 0 \text{ or } x < 10 \text{ and } y < 0)$ is same as $((x > 0 \text{ or } x < 10) \text{ and } y < 0)$

Your answer ABCD is incorrect



[Click here to show the correct answer](#)